# Assignment 3

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## Q1) Java Program to Multiply Two Matrices

```java
class MatrixMultiplication {
    public static void main(String args[]) {

        int a[][] = {
            {1, 2},
            {3, 4}
        };
        int b[][] = {
            {5, 6},
            {7, 8}
        };
        int c[][] = new int[2][2];

        for (int i = 0; i < 2; i++) {
            for (int j = 0; j < 2; j++) {
                c[i][j] = 0;
                for (int k = 0; k < 2; k++) {
                    c[i][j] = c[i][j] + a[i][k] * b[k][j];
                }
            }
        }
        System.out.println("Result Matrix:");
        for (int i = 0; i < 2; i++) {
            for (int j = 0; j < 2; j++) {
                System.out.print(c[i][j] + " ");
            }
            System.out.println();
        }
    }
}
```

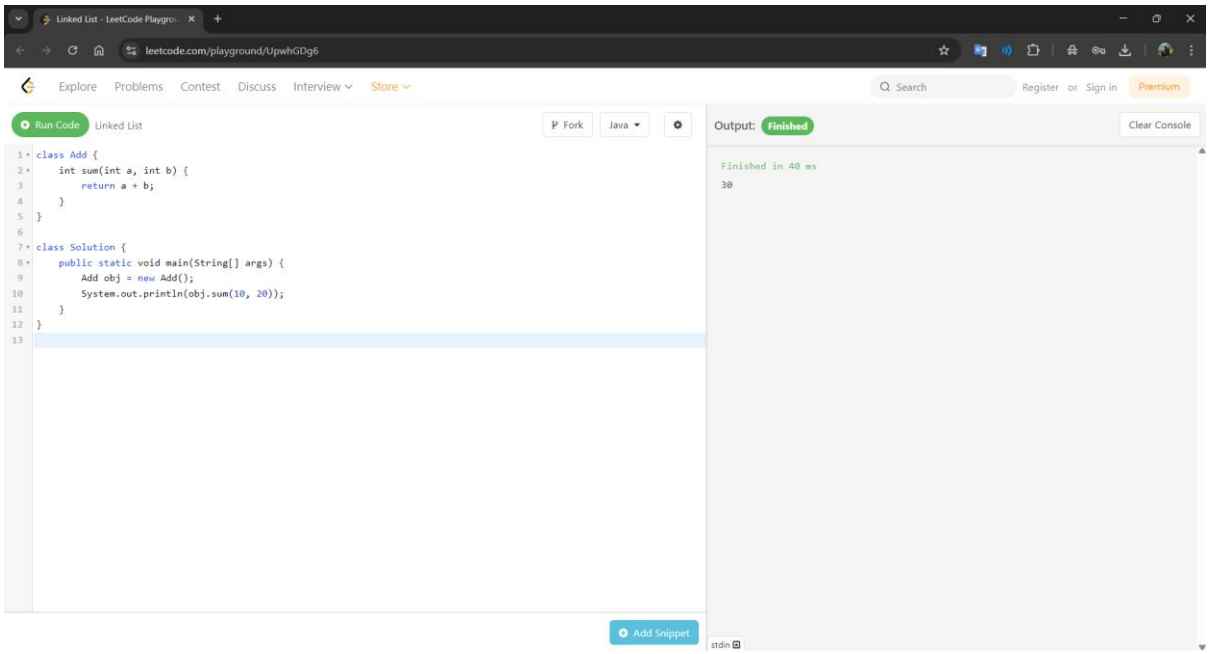
```
Finished in 61 ms
Result Matrix:
19 22
43 50
```

## Q2) Java Program to count the total number of vowels and consonants in a string

```java
class VowelConsonant {
    public static void main(String args[]) {

        String str = "Hello World";

        int vowels = 0;
        int consonants = 0;

        for (int i = 0; i < str.length(); i++) {
            char ch = str.charAt(i);

            if (ch >= 'A' && ch <= 'Z')
                ch = (char)(ch + 32);   // convert to lowercase

            if (ch >= 'a' && ch <= 'z') {
                if (ch == 'a' || ch == 'e' || ch == 'i' ||
                    ch == 'o' || ch == 'u')
                    vowels++;
                else
                    consonants++;
            }
        }

        System.out.println("Vowels = " + vowels);
        System.out.println("Consonants = " + consonants);
    }
}
```

```
Finished in 51 ms
Vowels = 3
Consonants = 7
```

Q3) Java program to call the function form one package to another package for any relevant problem.

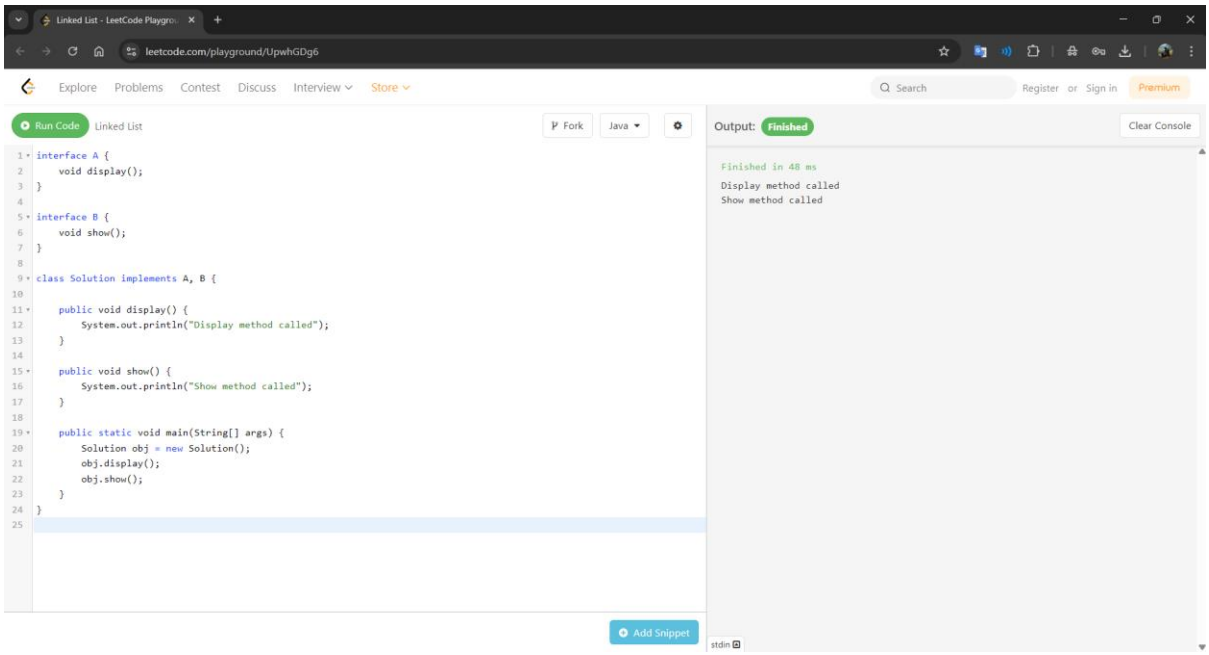


```java
class Add {
    int sum(int a, int b) {
        return a + b;
    }
}

class Solution {
    public static void main(String[] args) {
        Add obj = new Add();
        System.out.println(obj.sum(10, 20));
    }
}
```

Output: Finished

```
Finished in 40 ms
30
```

Q4) Java program to use interface with multiple inheritance for any relevant problem.



```java
interface A {
    void display();
}

interface B {
    void show();
}

class Solution implements A, B {

    public void display() {
        System.out.println("Display method called");
    }

    public void show() {
        System.out.println("Show method called");
    }

    public static void main(String[] args) {
        Solution obj = new Solution();
        obj.display();
        obj.show();
    }
}
```

Output: Finished

```
Finished in 48 ms
Display method called
Show method called
```